

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

Correct Answer: 66

Rationale

The correct answer is **66**. It's given that the right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. Since the diameter of the base of the cone is **6 cm**, the radius of the base is **3 cm**. The volume V , **in cm³**, of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, **in cm**, and r is the radius, **in cm**, of the base of the cone. Substituting **22** for h and **3** for r in this formula yields $V = \frac{1}{3}\pi(3)^2(22)$, or $V = 66\pi$. Therefore, the volume of the cone is **66π cm³**. It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **66**.